Huawei Access Node Failure SOP for YEL-ACC-52 at Yelahanka Sub-station (ALARM_6205)

Node: YEL-ACC-52 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_6205

Description

The Huawei Access Node YEL-ACC-52 at Yelahanka Sub-station has encountered a hardware/environmental fault resulting in an ASIC memory parity error, line card crash, or power supply redundancy loss, triggering a warning alarm (ALARM_6205). This failure domain is primarily attributed to the inherent vulnerability of the Access Node's Role-specific hardware design and environmental factors such as high temperature, humidity, and electromagnetic interference. The Line Card's ASIC memory parity error can be caused by faulty memory chips, incorrect configuration, or software bugs. The power supply redundancy loss might be due to a faulty power supply unit, incorrect configuration, or a software issue. The Huawei Access Node's control plane and data plane are designed to operate in tandem, with the control plane responsible for managing the data plane's operations. In this scenario, the diagnostic commands will focus on identifying the root cause of the failure, which could be a hardware, environmental, or software issue.

Pre-requisite Checks

- Verify the redundant link status of the Access Node.

- Check the environmental conditions at the Yelahanka Sub-station, specifically the temperature, humidity, and electromagnetic interference levels.

- Ensure the remote hands team has been dispatched to the Yelahanka Sub-station.

- Perform a visual inspection of the Access Node's hardware, focusing on the Line Card and power supply units.

Potential Root Causes

Cause	Probability
Faulty memory chips in the Line Card's ASIC	High
Incorrect configuration or software bugs	Medium
Faulty power supply unit	Low
Incorrect configuration or software issues	Medium
Environmental factors such as high temperature, humidity, or electromagnetic interference	Low

Diagnostic Commands & Expected Outputs

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Command: display this alarm 6205, Expected Output: ALARM_6205: ASIC memory parity error, Line Card crash, or Power supply redundancy loss, Description: This command will display the current alarm details,
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including the reason code, which will help identify the root cause of the failure.

Command: display hardware status, Expected Output: Hardware status: OK, but with warning, Description: This command will display the current hardware status, including any warnings or errors, which can help identify potential hardware issues.

Command: display power supply status, Expected Output: Power supply status: OK, but with warning, Description: This command will display the current power supply status, including any warnings or errors, which can help identify potential power supply issues.

Command: display environment status, Expected Output: Environment status: OK, but with warning, Description: This command will display the current environment status, including any warnings or errors, which can help identify potential environmental issues.
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Resolution Steps

Step 1: Step: Replace the faulty Line Card, Command: replace slot 1, Description: This step will replace the faulty Line Card with a new one, which should resolve the ASIC memory parity error issue.

Step 2: Step: Reset the power supply unit, Command: reset power supply 1, Description: This step will reset the power supply unit, which should resolve the power supply redundancy loss issue.

Step 3: Step: Adjust the environmental settings, Command: set environment temperature 20, Description: This step will adjust the environmental settings, such as temperature and humidity, to prevent future environmental issues.

Post-Check Verification

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Command: display alarm 6205, Expected Output: ALARM_6205: Cleared, Description: This command will verify that the alarm has been cleared, indicating that the issue has been resolved.

Command: display hardware status, Expected Output: Hardware status: OK, Description: This command will verify that the hardware status is OK, indicating that the issue has been resolved.

Command: display power supply status, Expected Output: Power supply status: OK, Description: This command will verify that the power supply status is OK, indicating that the issue has been resolved.
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Roll-back Procedures

- Step: Restore the original Line Card, Command: restore slot 1, Description: This step will restore the original Line Card, which should resolve any issues caused by the replacement.

- Step: Reset the power supply unit to its original state, Command: reset power supply 1 to original state, Description: This step will reset the power supply unit to its original state, which should resolve any issues caused by the reset.